

Testing During Design

Apart from the many advantages of testing during design, perhaps the most important is to test implementation of new technologies for which an ecosystem does not yet exist. Read on to find out how

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Testing during design helps in verification of sub-system level design, which permits fine-tuning to meet the system specification requirements. The designs that are tested during this stage are of individual parts of a final product, which means that they would be missing critical functionality that is required to make the system operable. This makes it important to simulate the other sub-components which are not yet available.

Alternatively, one could test subsets of functionality of the larger system that may be working at a given point in time, but not necessarily with all the functionality of the end product. "In industries like automotive and aerospace, where the end product is a complex system of systems, the ability to test subsystems through hardware-in-the-loop testing is a critical method to shorten the overall development cycle," explains Raviteja Chivukula, Technical Marketing Engineer, National Instruments.

How it is different from testing after design

Testing is done during the design process by the design engineers themselves. Here, their focus is to ensure that the electronics system design works in the real world just as it did on paper. They would typically be debugging the product to ensure that the intended functions are implemented one way or the other, and then optimise those features correctly.

On the other hand, testing after design is more focused on ensuring that the product has met the design specifications that were set by the customer.

Stephen Hire, general manager, Aeroflex Systems Pvt Ltd, says, "From an R&D perspective, testing during design entails two very different objectives for the two approaches. Testing during design is about enabling the engineer to make a design work, then debug it and finally optimise it. Testing after design is usually more concerned with proving that the design has met its design specifications."

How to successfully apply this methodology

The main point to understand and keep in mind is that the device under test (DUT) here is still not a complete system as it lacks many features. This requires the testing to be done in such a way that only certain parts of the system are considered, failing which the test would also invariably fail.

"The most optimal way to bring out issues is by doing unit testing of all the modules which go into a larger system. When systems get integrated, there is always a possibility of one corner case being missed, which is not handled properly during system level testing," explains Amit Sinha, CEO, Dexcel Electronics Designs.

From a functionality perspective, many things in the product might not be working. The goal is to plan testing to achieve measurement objectives when the system is not yet exhibiting high levels of functionality. A spokesman from Teledyne Lecroy adds, "One aspect of this challenge is to have a

good idea of the system's partitioning, whether in terms of subsystems or subsets of functionality that may involve multiple subsystems. Yet another aspect is to have an understanding of what the capabilities of the subsystem or subset should be, and devise a test to verify that functionality."

This can be explained by looking at cellular base stations as an example. During the earlier stages of a new technology's evolution, such as in the case of LTE-Advanced, there are no mobiles available that use the technology. "This requires design engineers to use test equipment as a test peer to help them make sure that the base station is correctly implementing LTE-A protocols and test features such as carrier ag-

